Completed MATLAB and Simulink results

CCEN 302 | Jesher | 2 November 2026

This is the current complete objective-results section. The earlier three-control report and previous nearest-source-time experiment are retained as history. All 36 conditions were generated and run independently in MATLAB and through the saved Simulink model, giving 72 actual result rows. No missing teammate output arrays were copied or relabeled as Simulink results.

Method

The source is `Audios/SpeechMusic.wav`: mono, 44,100 Hz, 16-bit PCM, 882,005 samples (20.000113379 s). The rate set is 44,100, 22,050, 11,025, 8,000 and 4,000 Hz, with effective bits 16, 8, 4 and 2. Native-rate cases have one route; each lower rate has filtered and unfiltered routes. Both platforms use the same settings.

The filtered route uses a Kaiser-window FIR with n=10 and beta=5 and rational conversion p/q. The unfiltered route selects every second/fourth sample at 22,050/11,025 Hz. At 8,000/4,000 Hz it uses linear interpolation at uniform output times without a dedicated anti-aliasing LPF, matching the sampling owner's handoff. Interpolation affects amplitude response. Output times stay within the source interval. No independent normalization is used. Current rows are linear_interp_v2; history/13fc434 preserves the previous nearest-source-time experiment.

The uniform signed quantizer uses step Delta=2/2^B, rounds to the nearest integer level, saturates its index to [-2^(B-1), 2^(B-1)-1], and multiplies by Delta. The 16-bit cases bypass further quantization as source/control cases. The visible Simulink implementation uses actual FIR Rate Conversion, Selector, Gain, Rounding Function and Saturation blocks. Linear interpolation is calculated inside the model using adjacent-source Selectors, Sum and Product blocks; processed MATLAB answers are not fed into the Simulink path.

Every processed array is exported at its actual output rate in a 16-bit PCM WAV. PCM conversion is explicitly nearest-integer rounding. The WAV is read back and reconstructed onto the original 44,100 Hz grid before subtracting. Each FIR block compensates its own known phase/delay; already compensated resample outputs and imported arrays receive no second shift. The full interval is retained: 882,005 samples, t=0 to 20.000090702948 s. Only deterministic end-grid rounding is discarded; there is no lag optimization, gain fitting or edge trimming.

`NMSE = sum((x-y).^2)/sum(x.^2)`; `SNR_dB = 10*log10(sum(x.^2)/sum((x-y).^2))`. The main table scores the **re-imported WAV**, while the CSV also records processing-array scores. Simulink displays calculate those processing-array scores inside the model. Positive-energy zero error gives NMSE=0/SNR=+Inf; silent-reference special cases are handled explicitly in `quality_metrics.m`.

Full condition table

Table 1. Aligned quality scores of all 36 conditions on each platform, measured after WAV export/re-import. Processing-array scores are separately recorded in Results/metrics.csv and platform_comparison.csv. Full precision and export conventions are retained in those CSVs.

Rate (Hz)	Bits	Filter	MATLAB NMSE	MATLAB SNR (dB)	Simulink SNR (dB)
44100	16	native	0	+Inf	+Inf
44100	8	native	0.00046095	33.3635	33.3635
44100	4	native	0.10316	9.8649	9.8649
44100	2	native	0.736785	1.32659	1.32659
22050	16	lpf	0.000807225	30.9301	30.9301
22050	8	lpf	0.00125268	29.0216	29.0216

Rate (Hz)	Bits	Filter	MATLAB NMSE	MATLAB SNR (dB)	Simulink SNR (dB)
22050	4	lpf	0.101621	9.93017	9.93017
22050	2	lpf	0.732912	1.34948	1.34948
22050	16	none	0.0012202	29.1357	29.1357
22050	8	none	0.0016631	27.7908	27.7908
22050	4	none	0.101983	9.91473	9.91473
22050	2	none	0.732973	1.34912	1.34912
11025	16	lpf	0.01302	18.8539	18.8539
11025	8	lpf	0.0134703	18.7062	18.7062
11025	4	lpf	0.112719	9.48003	9.48003
11025	2	lpf	0.731212	1.35957	1.35957
11025	16	none	0.0239321	16.2102	16.2102
11025	8	none	0.0243771	16.1302	16.1302
11025	4	none	0.123362	9.08817	9.08817
11025	2	none	0.736254	1.32973	1.32973
8000	16	lpf	0.0251219	15.9995	15.9995
8000	8	lpf	0.0255641	15.9237	15.9237
8000	4	lpf	0.124191	9.0591	9.0591
8000	2	lpf	0.734668	1.33909	1.33909
8000	16	none	0.0439778	13.5677	13.5677
8000	8	none	0.0444516	13.5211	13.5211
8000	4	none	0.143048	8.4452	8.4452
8000	2	none	0.743014	1.29003	1.29003
4000	16	lpf	0.122593	9.11536	9.11536
4000	8	lpf	0.123038	9.09962	9.09962
4000	4	lpf	0.218377	6.60792	6.60792
4000	2	lpf	0.784611	1.05345	1.05345
4000	16	none	0.224498	6.48788	6.48788
4000	8	none	0.225002	6.47813	6.47813
4000	4	none	0.31843	4.96986	4.96986
4000	2	none	0.834594	0.785249	0.785249

Maximum MATLAB/Simulink processing-sample difference across all pairs: **7.77156117238e-16** (tolerance 1e-10). Maximum re-imported WAV sample difference: **3.0517578125e-05**. Exported-WAV scores identical for all pairs: **False**. 2 pairs differ after export; tiny processing differences can fall on opposite sides of a PCM rounding boundary. The measured maximum exported-SNR difference is 1.61381049857e-07 dB. The export comparison allows at most one PCM step; raw processing arrays are checked independently at 1e-10. These results apply to this implementation and source.

Interpretation

At the native rate, reducing effective bits increases error and reduces SNR. For example, native 8-bit and 2-bit cases have SNRs of 33.3635 and 1.3266 dB. Filtering before rate reduction generally improves the high-precision sampling controls: at 4,000 Hz and 16-bit control precision, filtered/unfiltered SNRs are 9.1154/6.4879 dB. Low-bit quantization can dominate the total error, so the effect of filtering is smaller and need not be monotonic in every low-bit case. The approximate 6 dB/bit rule assumes a suitable ideal quantization-noise model; it is not imposed on this speech/music recording or on saturated low-bit levels.

Representative figures

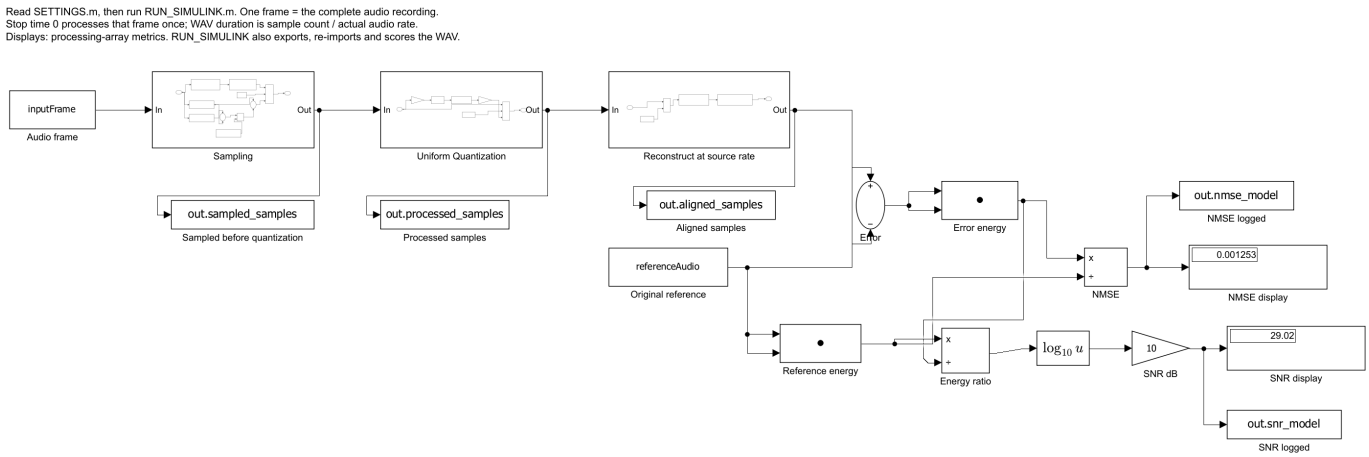


Figure 1. Saved `Audio_Demo.slx` processing diagram. Sampling, uniform quantization and reconstruction are explicit subsystems. The reference and reconstructed processing arrays feed Simulink energy calculations and NMSE/SNR displays. One frame is the complete recording; model stop time 0 evaluates that frame once and is not the WAV duration. MATLAB then writes and evaluates the WAV.

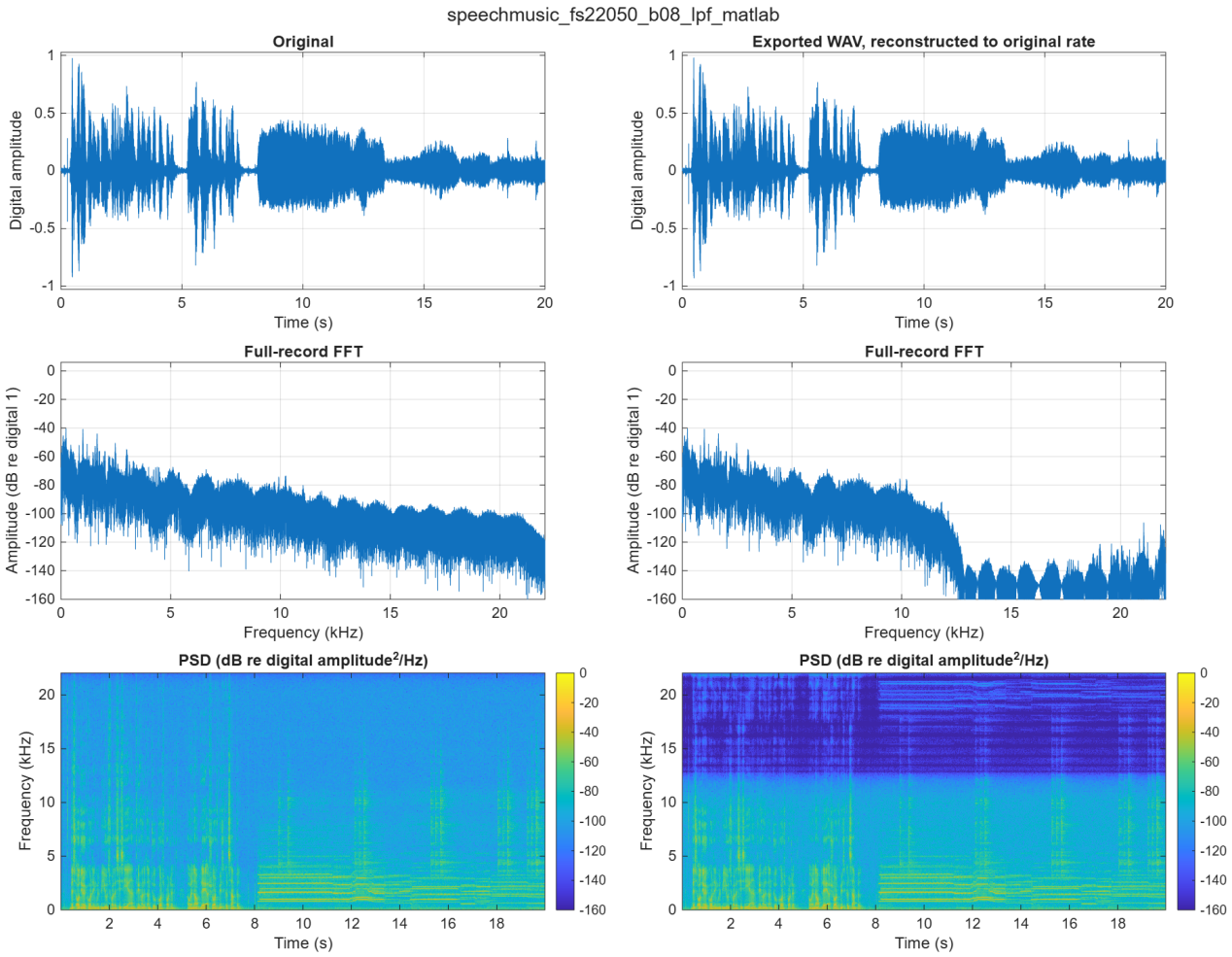


Figure 2. Combined 22,050 Hz, 8-bit filtered case: original (left) and re-imported WAV reconstructed to the original grid (right). Rows show waveform, full-record single-sided FFT magnitude and one-sided PSD spectrogram. The envelope and main low-frequency structure remain recognizable while higher frequencies are attenuated. NMSE=0.00125267620062; SNR=29.02161174 dB.

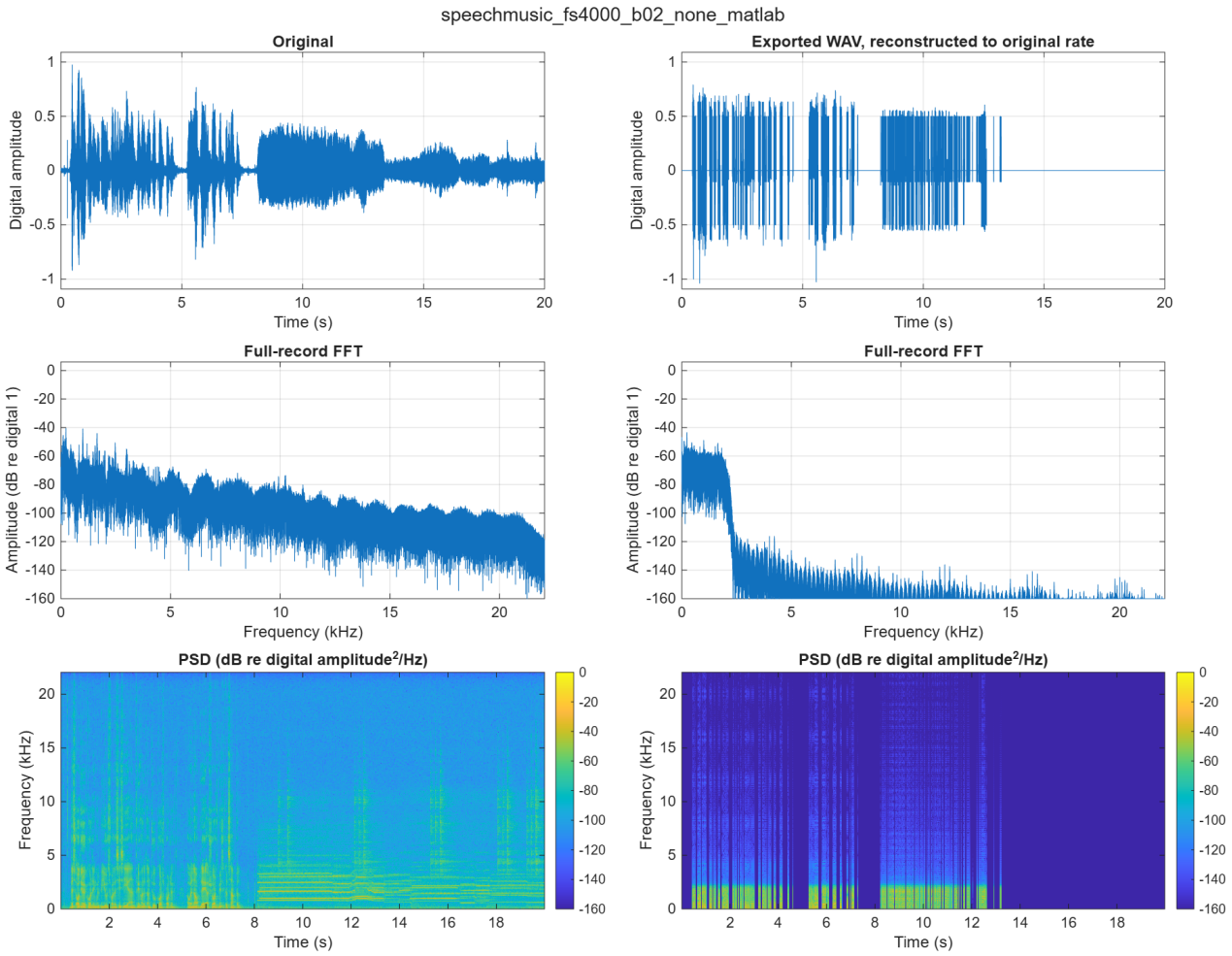


Figure 3. Combined 4,000 Hz, 2-bit unfiltered case with the same panel ordering as Figure 2. The two-bit sampled array uses at most four levels; the reconstructed trace interpolates those levels. Retained bandwidth is greatly reduced. Reconstruction can suppress images but cannot undo aliasing that occurred during unfiltered sampling. NMSE=0.834593657706; SNR=0.78524920 dB. Audible descriptors require actual listening.

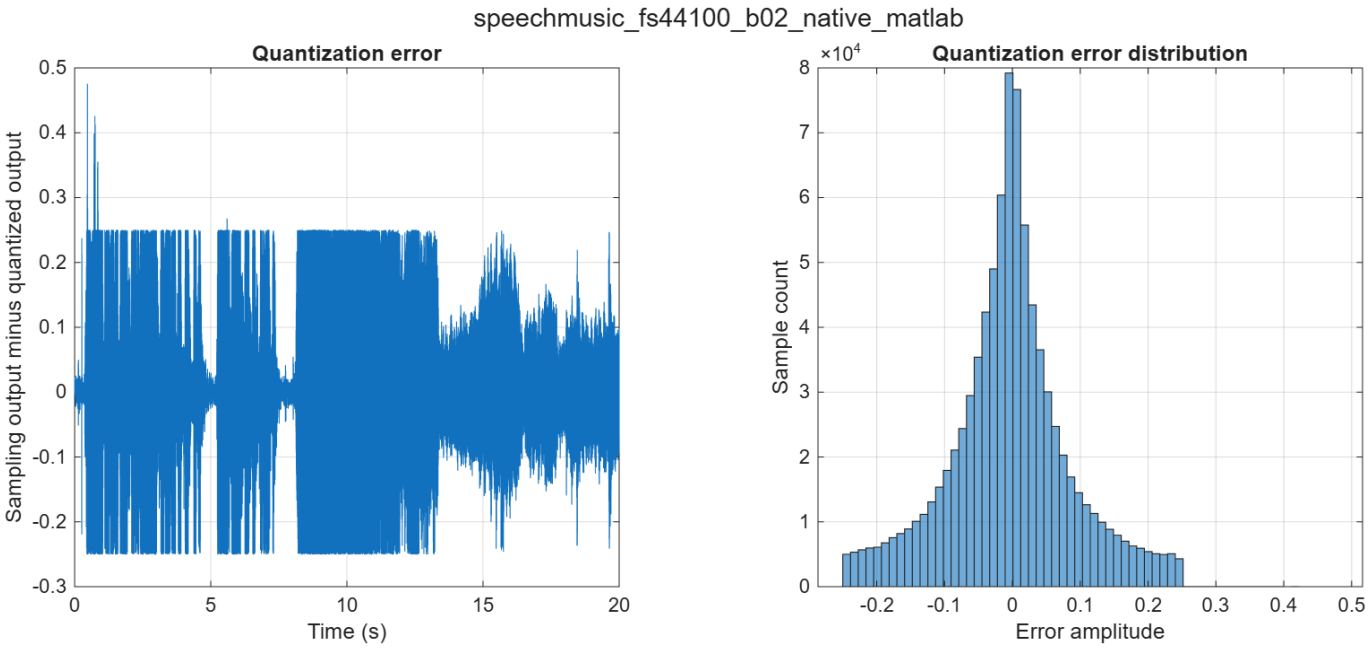


Figure 4. Native-rate 2-bit quantization error (sampling output minus quantized output) and its sample histogram. Low-bit saturation can produce errors larger than half a quantizer step; a uniform-noise distribution is not assumed. This plot isolates quantization error at the same time base.

The waveform/FFT display retains extrema in dense display bins for drawing speed; the full signal is used in every FFT and metric. Spectrogram settings for this source are periodic Hann 1,024 samples, 768 overlap, 2,048 DFT points, 44,100 Hz analysis rate, and PSD color limits -160 to 0 dB. FFT amplitudes are divided by the full

record length, paired positive-frequency bins are doubled, and a -160 dB display floor is used. These digital dB values are not calibrated sound-pressure levels.

Audio and storage

All 72 case WAVs passed actual rate, mono channel count, 16-bit container, sample-count and file-size checks. No export clipping occurred. There are 73 audio files in `Results/all_audio.zip`: the unchanged original plus 72 platform-labeled outputs. The accompanying 73-row audio manifest and every ZIP member were checked against the files byte for byte.

Table 2. Actual WAV sizes by rate (the four effective bit depths use the same 16-bit carrier). Ideal packed payload is `ceil(samples*effective_bits/8)` bytes for mono audio; it is distinct from these measured container sizes.

Rate (Hz)	Samples	Duration (s)	Actual WAV bytes
44100	882005	20.000113379	1764054
22050	441003	20.000136054	882050
11025	220502	20.000181406	441048
8000	160001	20.000125000	320046
4000	80001	20.000250000	160046

Each current WAV has 44 bytes beyond its two-byte-per-mono-sample PCM payload. Changing effective bits alone does not shrink these WAVs. Rate reduction changes the actual sample payload. Sample-count rounding accounts for the tiny differences in N/Fs durations; changing a WAV header alone is not rate conversion.

Listening and remaining human work

Jesher previously reported the original and the two earlier 22,050 Hz sampling controls as audible and clear. That observation is retained in `../listening/initial_observation.md`. It does not establish listening to the new 72 exports, especially the reduced-bit/combined cases. No new subjective observations, four-person comparison, speaker routing/speed confirmation, reviewer acceptance or rehearsal are invented here.

Numerical implementation, the full matrix, export/ZIP generation, plots and the live-demo entry points are complete with the available resources. The team still needs to listen to the new outputs, collect each person's notes with a fixed setup, confirm microphone provenance, review the chosen conventions and rehearse the live presentation. This self-contained demo was implemented here; it is not claimed as independently produced or accepted by another teammate.

Known-tone alias check

The optional `DEMONSTRATE_ALIASING.m` uses the existing 10 kHz sine recording and an 8 kHz output rate. The expected fold is `abs(10000-8000)=2000 Hz`. The measured dominant unfiltered output frequency was **2000.000 Hz**. The peak check omits 0.1 s at each edge only for tone-frequency inspection; all NMSE/SNR calculations still retain their full comparison intervals.

Exact sampling-handoff evaluation

The separate public evaluator imported the eight supplied MATLAB sampling arrays unchanged, plus the native reference. It reports 9 of 72 handoff cases available and 63 missing; it does not relabel the independent demo as teammate-produced output. Explicit top-level cases and nested samplingHandoff.cases are accepted, with explicit IDs taking precedence over fallback controls and duplicate explicit IDs rejected. Method, producer filter/timing metadata and artifact provenance are retained.

These b16 controls retain double-valued sampling results without an added quantizer. Flags: quantizationApplied=false, normalizationApplied=false, delayCompensated=true, alignmentPolicy=full_record_resample_v1. Unfiltered routes have no FIR delay; filtered routes have already been compensated. No second correction is applied.

Table 3. Sampling controls measured from supplied processing doubles and from the handoff evaluator's own WAV export/re-import. Both evaluators now use explicit nearest-integer PCM16 conversion, with half ties away from zero and measured error at most half a PCM step. This replaces the legacy implicit double-input audiowrite conversion; historical WAV metrics preserve the older behavior. Effective reduced-bit quantization is unchanged. Compare matching representations.

Rate (Hz)	Filter	Processing NMSE	Processing SNR (dB)	Exported SNR (dB)
44100	native	0	+Inf	+Inf
22050	lpf	0.000807222	30.9301	30.9301
22050	none	0.0012202	29.1357	29.1357
11025	lpf	0.01302	18.8539	18.8539
11025	none	0.0239321	16.2102	16.2102
8000	lpf	0.0251219	15.9995	15.9995
8000	none	0.0439778	13.5677	13.5677
4000	lpf	0.122593	9.11536	9.11536
4000	none	0.224498	6.48788	6.48788

Integration and timing checks

All 22 MATLAB regression tests and all 20 existing independent Python tests passed. Eight imported arrays match the supplied arrays exactly. All eight regenerated MATLAB controls agree with their supplied counterparts within 1e-12. Sixteen 300 Hz checks (four rates, both filter modes, both actual platforms) retain all 882,000 sample times of that tone recording and have zero diagnostic residual lag; the measured dominant frequency is within 1 Hz of 300 Hz. Diagnostic lag is never applied to the measurements.

Four actual-platform alias checks cover 10 kHz input at 11,025 and 8,000 Hz. The measured aliases are within 1 Hz of 1,025 and 2,000 Hz respectively. Filtered interior RMS is more than 40 dB below the unfiltered path in each check. Only tone-frequency diagnostics omit 0.1 s at each edge; every NMSE/SNR score retains the full reference interval and its reconstruction boundary errors.

This integration does not complete Abdulrahman's separate Simulink deliverables or Khaled's quantizer/storage work. Their implementations and checklists remain unchanged.

Table 4. Actual-platform 10 kHz tone checks on processing arrays. RMS reduction compares filtered and unfiltered interiors; this is not an acoustic SNR.

Platform	Rate (Hz)	Expected alias (Hz)	Measured alias (Hz)	Filtered RMS reduction (dB)
matlab	11025	1025	1025.004696	68.657
simulink	11025	1025	1025.004696	68.657
matlab	8000	2000	2000.000000	76.059
simulink	8000	2000	2000.000000	76.059